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Tests of Covariance Matrices for High Dimensional Multivariate Data Under Non Normality

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Ahmad and von Rosen (2014) presented test statistics for sphericity and identity of the covariance matrix of a multivariate normal distribution when the dimension, p, exceeds the sample size, n. In this note, we show that their statistics are robust to normality assumption, when normality is replaced with certain mild assumptions on the traces of the covariance matrix. Under such assumptions, the test statistics are shown to follow the same asymptotic normal distribution as under normality for large p, also when p >> n. The asymptotic normality is proved using the theory of U-*statistics, and is based on very general conditions, particularly avoiding any relationship between n and p.*

Keywords Non normality; High dimensionality; Sphericity.

Mathematics Subject Classification 62H15.

1. Introduction

Let

$$\mathbf{X}_k = (X_{k1}, \ldots, X_{kp})' \sim \mathcal{F}, \tag{1}$$

$k = 1, \ldots, n$, be n independent and identically distributed random vectors, where $\mathcal{F}$ denotes certain non degenerate p-variate distribution. Let $\mathrm{E}(\mathbf{X}_k) = \boldsymbol{\mu}$ and $\mathrm{Cov}(\mathbf{X}_k) = \boldsymbol{\Sigma}$, $\boldsymbol{\Sigma} > 0$, denote the mean vector and the covariance matrix of $\mathbf{X}_k$, respectively. We are interested to test the hypotheses

$$H_{01} : \boldsymbol{\Sigma} = \sigma^2\mathbf{I} \qquad \text{vs} \qquad H_{11} : \boldsymbol{\Sigma} \neq \sigma^2\mathbf{I}$$

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and

$$H_{02} : \mathbf{\Sigma} = \mathbf{I} \qquad \text{vs} \qquad H_{12} : \mathbf{\Sigma} \neq \mathbf{I},$$

when $p > n$, where $\mathbf{I}$ is the identity matrix, and $\sigma^2 > 0$ is some constant. Here, H_{01} refers to the well-known sphericity hypothesis whereas H_{02} is a convenient representation of a more general hypothesis $\mathbf{\Sigma} = \mathbf{\Sigma}_0$, where $\mathbf{\Sigma}_0$ is any known positive definite covariance matrix. When $\mathcal{F}$ is multivariate normal and $n > p$, the asymptotically locally most powerful invariant tests for H_{01} and H_{02}, as developed by John (1971) and Nagao (1973), are

$$U = \frac{1}{p}\text{tr}\left(\frac{\mathbf{S}}{\frac{1}{p}\text{tr}(\mathbf{S})} - \mathbf{I}\right)^2, \qquad \text{and} \qquad V = \frac{1}{p}\text{tr}(\mathbf{S} - \mathbf{I})^2, \tag{2}$$

respectively, where $\mathbf{S}$ is the sample estimator of $\mathbf{\Sigma}$. The construction of U and V is based on using $\mathbf{S}$ as a plug-in estimator of $\mathbf{\Sigma}$ in

$$\frac{1}{p}\text{tr}\left(\frac{\mathbf{\Sigma}}{\frac{1}{p}\text{tr}(\mathbf{\Sigma})} - \mathbf{I}\right)^2 = \frac{\frac{1}{p}\text{tr}(\mathbf{\Sigma}^2)}{\left[\frac{1}{p}\text{tr}(\mathbf{\Sigma})\right]^2} - 1 = \frac{p\text{tr}(\mathbf{\Sigma}^2)}{[\text{tr}(\mathbf{\Sigma})]^2} - 1 \tag{3}$$

and

$$\frac{1}{p}\text{tr}(\mathbf{\Sigma} - \mathbf{I})^2 = \frac{1}{p}\text{tr}(\mathbf{\Sigma}^2) - \frac{2}{p}\text{tr}(\mathbf{\Sigma}) + 1, \tag{4}$$

respectively. For $\mathbf{X} \sim \mathcal{N}_p(\boldsymbol{\mu}, \mathbf{\Sigma})$, Ahmad and von Rosen (2014) presented the following test statistics as modified versions of (3) and (4) that can be validly used to test H_{01} and H_{02} even when $p > n$.

$$T_1 = \frac{pE_3}{E_2} - 1 \tag{5}$$

$$T_2 = \frac{1}{p}E_3 - \frac{2}{p}E_1 + 1. \tag{6}$$

Here, E_1, E_2, E_3 are the estimators of $\text{tr}(\mathbf{\Sigma})$, $[\text{tr}(\mathbf{\Sigma})]^2$, and $\text{tr}(\mathbf{\Sigma}^2)$, defined as

$$E_1 = \frac{1}{n}\sum_{k=1}^{n} A_k, \tag{7}$$

$$E_2 = \frac{1}{n(n-1)} \underbrace{\sum_{k=1}^{n}\sum_{l=1}^{n}}_{k \neq l} A_k A_l, \tag{8}$$

$$E_3 = \frac{1}{n(n-1)} \underbrace{\sum_{k=1}^{n}\sum_{l=1}^{n}}_{k \neq l} A_{kl}^2, \tag{9}$$

where $A_k = \mathbf{X}_k'\mathbf{X}_k$ is a quadratic form, and $A_{kl} = \mathbf{X}_k'\mathbf{X}_l$, $k \neq l$, is a symmetric bilinear form. They show that E_1, E_2, and E_3 are unbiased and consistent estimators of the corresponding traces, and the consistency is unaffected by a very large dimension, even larger than the sample size. Based on these properties of trace estimators, they show, using the theory of one-sample non degenerate U-statistics, that T_1 and T_2 are asymptotically normal when both $n, p \to \infty$. In particular, under H_{0i}, T_i is proved asymptotically $N(0, 4)$, $i = 1, 2$. The high dimensional asymptotics are not based on any strict assumption on the traces of $\mathbf{\Sigma}$, or any relationship between n and p. The extensive simulation results show the accuracy of T_1 and T_2 under very general conditions.

Recently, there have been several other test statistics for H_{01} and H_{02} presented in the literature the asymptotic distributions of which are derived under different settings. Ledoit and Wolf (2002) originally investigated the test statistics given in (2) for their use under high-dimensional setting assuming normality. Under several assumptions on the moments of the eigenvalues of the covariance matrix, and additionally, assuming $p/n \to c \in (0, \infty)$, they derive the asymptotic normal distribution of the test statistics. Without assuming normality, but under other assumptions relating the covariance matrices, Chen et al. (2010) derived test statistics for the same two null hypotheses. Similar attempts, with and without normality assumptions, are carried out by Srivastava (2005) and Srivastava et al. (2011), respectively. For a more detailed review of literature, along with a discussion on frequently used assumptions for high-dimensional inference about covariance matrices, see Ahmad and von Rosen (2014).

In this article, we show that T_1 and T_2, as given in Eqs. (5) and (6), are robust to normality assumption, and can still be used to test H_{01} and H_{02} when $p > n$. The asymptotic distributions of both test statistics are shown to be normal, particularly following the same null distributions, $N(0, 4)$, as under normality. We take full advantage of the non parametric nature of the U-statistics theory to show asymptotics of the test statistics when we relax normality. The normality assumption is replaced with certain inevitable moment assumptions on the underlying multivariate model, and a few mild assumptions on $\mathbf{\Sigma}$.

The next section presents the main results of the paper. We first show that the properties of trace estimators, E_1, E_2, and E_3, are intact even when normality is relaxed. Then, using these properties and the theory of U-statistics, we show the asymptotic normality of T_1 and T_2 when both $n \to \infty$ and $p \to \infty$. These results, proved in the appendix, are the non normal counterparts of Lemma 2.2, and Theorems 2.4 and 2.6 (along with their corollaries) in Ahmad and von Rosen (2014).

2. The test statistics

Since, the asymptotic properties of T_1 and T_2 mostly rest on those of the moment estimators of the three traces, i.e., E_1, E_2, and E_3, we first discuss the properties of these estimators. For this, we need to define the underlying multivariate model, Model (1), more precisely, and set some assumptions. For Model (1), define

$$\mathbf{X}_k = \mathbf{\Lambda}\mathbf{Y}_k, \tag{10}$$

where $\mathbf{Y}_k = (Y_{k1}, \ldots, Y_{kp})'$ such that $\mathrm{E}(\mathbf{Y}_k) = \mathbf{0}$ and $\mathrm{Cov}(\mathbf{Y}_k) = \mathbf{I}$, and $\mathbf{\Lambda}_p$, is a known matrix of constants such that $\mathbf{\Lambda}'\mathbf{\Lambda} = \mathbf{A}$, and $\mathbf{\Lambda}\mathbf{\Lambda}' = \mathbf{\Sigma} > 0$. The set up of Model (10) is very similar to the one proposed in Bai and Saranadasa (1996) to develop a two-sample mean testing statistic for high dimensional data. However, we will compute the asymptotic distribution of the test statistics under relatively milder assumptions than those required by Bai and Saranadasa model. Moreover, this asymptotic limit will be derived by using the theory of U-statistics which, by their properties, provide minimum variance unbiased estimators (Serfling, 1980, p. 176; Polansky, 2011, p. 482). This helps us establish that the estimators, E_1, E_2, E_3, used to construct the test statistics, are efficient, eventually leading to a better convergence of the statistic to its limit distribution. For a similar use of asymptotic theory of U-statistics under non normality, for one- and two-sample mean testing cases, see Ahmad et al (2012) and Ahmad (2014), respectively.

In the following, we state the assumptions needed to arrive at the asymptotic limit. We begin by assuming that the elements of $\mathbf{Y}_k$ are independent and identically distributed with finite fourth moment stated as

Assumption 2.1. *$E(Y_{ks}^4) = 3 + \gamma$, where γ is a finite constant.*

Assumption 2.2. *As $p \to \infty$, $\frac{tr(\mathbf{\Sigma}^i \odot \mathbf{\Sigma}^j)}{tr(\mathbf{\Sigma}^i \otimes \mathbf{\Sigma}^j)} \to 0$, $i, j = 1, 2, 3$, $i + j \leq 4$, where. $\odot$ denotes the Hadamard product and $\otimes$ denote the Kronecker product.*

Assumption 2.1 is essential. It replaces normality assumption, and is needed since moments of quadratic and bilinear forms will involve fourth moment of the random variables. Assumption 2.2 is milder and more practical than is usually adopted in the literature. The assumption is trivially satisfied when $\mathbf{\Sigma} = \mathbf{I}$. Otherwise, let $\mathbf{\Sigma}$ be compound symmetric, one of the most commonly used covariance structures (Wolfinger, 1996). Precisely, let $\mathbf{\Sigma} = \sigma^2[(1 - \rho)\mathbf{I} + \rho\mathbf{J}]$ where $\mathbf{J}$ is the matrix of 1's, and $\sigma^2 > 0$ and ρ are constants, $-1/(p - 1) \leq \rho \leq 1$. Assuming $\rho = 0.5$, $\sigma^2 = 1$, so that $\mathbf{\Sigma} > 0$, the validity of Assumption 2.2 can be immediately verified. For example, for $i = j = 1, 2, 3$, the ratio of the traces in the assumption is $O(1/p)$, verifying the assumption.

Based on these assumptions, the properties of the estimators, E_1, E_2, E_3, are proved in Appendix B (see Lemma 2.3 below). Note that, although the proof can be easily established using the first principles on the sums of independent and identically distributed random vectors, the proof in the appendix uses the fact that the estimators E_1, E_2, E_3, naturally appear in the form of U-statistics (Hoeffding, 1948). This relieves us of many extra computations and arguments needed for the asymptotic normality of the test statistics. For further reference, we here give a brief outline of a one-sample U-statistics, introducing the basic components used to prove the main results in the Appendix. More details can be found in Lehmann (1999, Ch. 6), Koroljuk and Borovskich (1994), Lee (1990), and Serfling (1980, Ch. 5).

For a symmetric function, $h(X_1, \ldots, X_m) : \mathcal{R}^m \to \mathcal{R}$ with $\mathrm{E}(h(.)) = \theta$, a one-sample U-statistic is defined as

$$U_n = \binom{n}{m}^{-1} \sum_{1 \leq i_1 < \ldots i_m \leq n} h(X_{i_1}, \ldots, X_{i_m}), \tag{11}$$

where the summation encompasses all possible permutations of $X_1, \ldots, X_m$, and $\mathrm{E}(U_n) = \theta$. The elements, $X_1, \ldots, X_m$ are assumed independent and identically distributed and may refer to random variables or, as for our case, random vectors. Note that each sum in U_n is composed of independent elements, which makes the unbiasedness of U_n self-evident, but since different sums of elements share common indices, the sums under different permutations in U_n are not independent which makes the computation of variance (and higher moments) cumbersome. The same problem persists in establishing the asymptotic normality of U_n. To compute the variance of U_n, this problem is countered by counting the possible cases which share, say c, common elements, and then summing up all such variances since the elements are identically distributed. Define

$$h_c(x_1, \ldots, x_c) = \mathrm{E}(h(X_1, \ldots, X_m)|X_1 = x_1, \ldots, X_c = x_c),$$

$1 \leq c \leq m$, with $h_m(\cdot) = h(\cdot)$, and let $\xi_c = \mathrm{Var}(h_c(x_1, \ldots, x_c))$, $c = 1, \ldots, m$. Then, the variance of U_n can be written as

$$\mathrm{Var}(U_n) = \binom{n}{m}^{-1} \sum_{c=1}^{m} \binom{m}{c} \binom{n-m}{m-c} \xi_c. \tag{12}$$

A very useful property of the U-statistics, concerning Eq. (12), is their optimality, in that they have the minimum variance in the class of unbiased estimators (see Polansky, 2011, Theorem 11.3, p. 482). Finally, to establish the asymptotic normality of U_n, the problem of correlated sums is pacified by projecting U_n on to a space of functions which are composed of the sums of independent random variables and are asymptotically equivalent to U_n. For a nice exposition on the formulation of these projections and their use in deriving asymptotic normality of U_n, see Polansky (2011, Ch. 11). Then, for $0 < \xi_c < \infty$, it can be shown that (see also Lehmann, 1999, Theorem 6.1.2(ii), p. 369)

$$\frac{U_n - \mathrm{E}(U_n)}{\mathrm{Var}(U_n)} \xrightarrow{\mathcal{D}} N(0, 1), \tag{13}$$

where $\xrightarrow{\mathcal{D}}$ denotes convergence in distribution, and N(0,1) denotes the standard normal distribution.

Now, suppose we have two such U-statistics, say $U_{n,1}$ and $U_{n,2}$ with their respective symmetric kernels, $h_1(X_1, \ldots, X_{m_1})$ and $h_2(X_1, \ldots, X_{m_2})$, respectively. Define the conditional expectations, $h_c(\cdot)$, for both kernels, same as explained for the one-sample case above, and $\xi_{cc} = \mathrm{Cov}(h_{1,c}(X_1, \ldots, X_c), h_{2,c}(X_1, \ldots, X_c))$ as the covariance between $h_1(\cdot)$ and $h_2(\cdot)$ when they share c elements in common. Then, assuming $m_1 \leq m_2$, the covariance between $U_{n,1}$ and $U_{n,2}$ can be computed by counting the covariances, ξ_{cc} for all $c = 1, \ldots, m_1$, such that

$$\mathrm{Cov}(U_{n,1}, U_{n,2}) = \binom{n}{m_1}^{-1} \sum_{c=1}^{m_1} \binom{m_2}{c} \binom{n-m_2}{m_1-c} \xi_{cc}. \tag{14}$$

For the estimators, E_1, E_2, and E_3, as defined in Eqs. (7)–(9), we have $m = 1, 2$ and 2, respectively, with their corresponding symmetric kernels

$$h(\mathbf{X}_k) = \frac{1}{\mathrm{tr}(\mathbf{\Sigma})} A_k, \quad h(\mathbf{X}_k, \mathbf{X}_l) = \frac{1}{[\mathrm{tr}(\mathbf{\Sigma})]^2} A_k A_l, \quad h(\mathbf{X}_k, \mathbf{X}_l) = \frac{1}{\mathrm{tr}(\mathbf{\Sigma}^2)} A_{kl}^2, \tag{15}$$

where $A_k = \mathbf{X}_k' \mathbf{X}_k$, $A_l = \mathbf{X}_l' \mathbf{X}_l$, and $A_{kl} = \mathbf{X}_k' \mathbf{X}_l$, $k \neq l$. Then, using these kernels and the results on the moments of U-statistics as explained above, the following lemma on the properties of estimators is proved in B.

Lemma 2.1. *Given the estimators E_1, E_2, E_3 as in Eqs. (7)-(9). Then*

$$E(E_1) = tr(\mathbf{\Sigma}), \qquad E(E_2) = [tr(\mathbf{\Sigma})]^2, \qquad E(E_3) = tr(\mathbf{\Sigma})^2.$$

Further, under Assumption 2.1

$$Var(E_1) = \frac{2}{n} tr(\mathbf{\Sigma}^2) + \gamma tr(\mathbf{A} \odot \mathbf{A}) \tag{16}$$

$$\begin{aligned} Var(E_2) = \frac{8}{n(n-1)} \Bigg[&(n-1) tr(\mathbf{\Sigma}^2)[tr(\mathbf{\Sigma})]^2 + [tr(\mathbf{\Sigma}^2)]^2 + \frac{\gamma^2}{4}[tr(\mathbf{A} \odot \mathbf{A})]^2 \\ &+ \gamma tr(\mathbf{A} \odot \mathbf{A}) \left\{ tr(\mathbf{\Sigma}^2) + \frac{n-1}{2}[tr(\mathbf{\Sigma})]^2 \right\} \Bigg] \end{aligned} \tag{17}$$

$$\begin{aligned} Var(E_3) = \frac{4}{n(n-1)} \Bigg[&(2n-1) tr(\mathbf{\Sigma}^4) + [tr(\mathbf{\Sigma}^2)]^2 + \gamma (n+1) tr(\mathbf{A}^2 \odot \mathbf{A}^2) \\ &+ \frac{\gamma^2}{2} \sum_{i=1}^{p} \sum_{j=1}^{p} A_{ij}^4 \Bigg] \end{aligned} \tag{18}$$

$$Cov(E_1, E_2) = \frac{4}{n} \left[tr(\mathbf{\Sigma}) tr(\mathbf{\Sigma}^2) + \frac{\gamma}{2} tr(\mathbf{\Sigma}) tr(\mathbf{A} \odot \mathbf{A}) \right] \tag{19}$$

$$Cov(E_1, E_3) = \frac{4}{n} \left[tr(\mathbf{\Sigma}^3) + \frac{\gamma}{2} tr(\mathbf{A}^2 \odot \mathbf{A}) \right] \tag{20}$$

$$\begin{aligned} Cov(E_2, E_3) = \frac{8}{n(n-1)} \Bigg[&tr(\mathbf{\Sigma}^4) + (n-1) tr(\mathbf{\Sigma}) tr(\mathbf{\Sigma}^3) + \frac{\gamma}{2}(n-1) tr(\mathbf{\Sigma}) tr(\mathbf{A}^2 \odot \mathbf{A}) \\ &+ \gamma tr(\mathbf{A}^3 \odot \mathbf{A}) + \frac{\gamma}{4} tr(\mathbf{A} \odot \mathbf{ADA}) \Bigg] \end{aligned} \tag{21}$$

where $\mathbf{D} = diag(\mathbf{A})$.

Following corollary is an immediate consequence of Lemma 2.3 and will be very useful in the proof of the main theorem (Theorem 2.5).

Corollary 2.1. *Given Lemma 2.1. Then,*

$$Cov\left(\frac{E_i}{E(E_i)}, \frac{E_j}{E(E_j)}\right) \leq O\left(\frac{1}{n}\right),$$

i.e., the variance-ratios ($i = j$) and the covariance-ratios ($i \neq j$) are uniformly bounded in the dimension, p, *for every* n.

As can be witnessed from Appendix A, the proofs of Lemma 2.1 and Corollary 2.1 are mainly facilitated by the norming of the kernels of the U-statistics by the corresponding traces; see (15). This helps us control the behavior of the kernels when $p \to \infty$, in particular, ensuring the square-integrability of the kernels, i.e., $\mathrm{E}(h^2(\cdot)) < \infty$ for each $h(\cdot)$. Moreover, as will be shown in the proof of the main theorem (Theorem 2.5), it leads to a non degenerate limit of the test statistic.

It must be noted that, the proof of the Corollary 2.4 does not require that $n \to \infty$, rather the variances and covariances are bounded only for $p \to \infty$. As will be made more clear in the proof of the asymptotic normality of the test statistics, the validity of the bounds in Corollary 2.1 for any n clues to the fact that the estimators, and hence the test statistics, can be used for a very large p and moderate n. Clearly, to establish consistency of the estimators by showing the vanishing of their variance- and covariance-ratios, we need to set $n \to \infty$. Similarly, following the asymptotic theory of a U-statistic, the condition that $n \to \infty$ is an inevitable requirement to complete the proof of asymptotic normality of the test statistics. But, apart from being a mathematical connotation, the accuracy of the test statistics, for practical purposes, does not seriously depend on the largeness of n.

The asymptotic normality of T_1 and T_2, under normality, is shown in Ahmad and von Rosen (2014) using the theory of U-statistics. But, being essentially a non parametric technique, the asymptotic theory of U-statistics does not require normality assumption. This fact, along with the uniform boundedness of variances and covariances of trace estimators in p, following Lemma 2.1, helps us a great deal to fix the asymptotic normality of the test statistics. We proceed as follows.

Consider T_1 as in Eq. (5). Define $\psi = \frac{p\mathrm{tr}(\boldsymbol{\Sigma}^2)}{[\mathrm{tr}(\boldsymbol{\Sigma})]^2}$, so that

$$T_1 = \frac{pE_3}{E_2} - 1 = \widehat{\psi} - 1,$$

where $\widehat{\psi} = \frac{pE_3}{E_2}$ is an estimator of ψ. Then, we can write

$$\frac{T_1 + 1}{\psi} = \frac{\mathrm{E}_3/\mathrm{tr}(\boldsymbol{\Sigma}^2)}{\mathrm{E}_2/[\mathrm{tr}(\boldsymbol{\Sigma})]^2}. \tag{22}$$

Similarly, for T_2, as defined in Eq. (6), we have

$$T_2 - \frac{1}{p}\mathrm{tr}(\boldsymbol{\Sigma} - \mathbf{I})^2 = \frac{\mathrm{tr}(\boldsymbol{\Sigma}^2)}{p}\left(\frac{\mathrm{E}_3}{\mathrm{tr}(\boldsymbol{\Sigma}^2)} - 1\right) - \frac{2\mathrm{tr}(\boldsymbol{\Sigma})}{p}\left(\frac{\mathrm{E}_1}{\mathrm{tr}(\boldsymbol{\Sigma})} - 1\right). \tag{23}$$

Note that, T_1 is a non linear function of the random variables, E_2 and E_3, and T_2 is a linear function of the random variables E_1 and E_3. This complicates the computations of the moments of the test statistics. But, using the results of Lemma 2.1, it can be immediately shown that

$$\frac{\mathrm{E}_1}{[\mathrm{tr}(\boldsymbol{\Sigma})]} \xrightarrow{\mathcal{P}} 1, \quad \text{and} \quad \frac{\mathrm{E}_2}{[\mathrm{tr}(\boldsymbol{\Sigma})]^2} \xrightarrow{\mathcal{P}} 1, \tag{24}$$

under Assumption 2.2, where $\xrightarrow{\mathcal{P}}$ denotes convergence in probability. Then, Eqs. (22) and (23) reduce to the following simple forms:

$$\frac{T_1 + 1}{\psi} - 1 = U_n - 1 \tag{25}$$

$$\frac{pT_2 - \mathrm{tr}(\boldsymbol{\Sigma} - \mathbf{I})^2}{\mathrm{tr}(\boldsymbol{\Sigma}^2)} = U_n - 1, \tag{26}$$

where

$$U_n = \frac{1}{n(n-1)} \sum_{k=1}^{n} \sum_{\substack{l=1 \\ k \neq l}}^{n} \frac{A_{kl}^2}{\mathrm{tr}(\boldsymbol{\Sigma}^2)} \tag{27}$$

is a one-sample U-statistic (Hoeffding, 1948; Lehmann, 1999, Ch. 6). Therefore, the asymptotic normality of both T_1 and T_2 follows from the asymptotic normality of a one-sample (non degenerate) U-statistic. Our proofs closely follow the assertions of Serfling (1980, Ch. 5) and Lehmann (1999, Theorem 6.1.2(ii), p. 369). The following theorem summarizes the results.

Theorem 2.1. *Let T_1 and T_2 be as defined in Eqs. (5) and (6), and consider their formulations in terms of U-statistics as given in Eqs. (25) and (26), respectively. Then, under Assumptions 2.1 and 2.2*

$$\sigma_{T_1}^{-1}\left(\frac{T_1+1}{\psi}-1\right) \xrightarrow{\mathcal{D}} N(0,1), \tag{28}$$

$$\sigma_{T_2}^{-1}\left[T_2-\frac{1}{p}tr(\mathbf{\Sigma}-\mathbf{I})^2\right] \xrightarrow{\mathcal{D}} N(0,1), \tag{29}$$

as $n, p \to \infty$, where $\sigma_{T_1}^2$ and $\sigma_{T_2}^2$ denote the variances of the tests statistics. In particular, under H_{0i},

$$\frac{n}{2}T_i \xrightarrow{\mathcal{D}} N(0,1), \quad i=1,2.$$

Proof. From Lemma 2.1, $\mathrm{E}[\mathrm{E}_1/\mathrm{tr}(\mathbf{\Sigma})]=1$, $\mathrm{E}[\mathrm{E}_2/[\mathrm{tr}(\mathbf{\Sigma})]^2]=1$, and, under Assumption 2.2, $\mathrm{Var}[\mathrm{E}_1/\mathrm{tr}(\mathbf{\Sigma})] \le 2/n$, and $\mathrm{Var}[\mathrm{E}_1/\mathrm{tr}(\mathbf{\Sigma})] \le 16/(n-1)$. Then, as $n, p \to \infty$, $\mathrm{E}_1/\mathrm{tr}(\mathbf{\Sigma}) \to \mathcal{P}1$ and $\mathrm{E}_2/[\mathrm{tr}(\mathbf{\Sigma})]^2 \to \mathcal{P}1$, by Chebychev's inequality. We then have to show the asymptotic normality of U_n in Eq. (27), and for this we need to compute $\mathrm{Var}(U_n)$ using Eq. (12). From the proof of Lemma 2.1, we note that

$$\xi_1 = \frac{2\mathrm{tr}(\mathbf{\Sigma}^4)}{[\mathrm{tr}(\mathbf{\Sigma}^2)]^2} + \gamma\frac{\mathrm{tr}(\mathbf{A}^2\odot\mathbf{A}^2)}{[\mathrm{tr}(\mathbf{\Sigma}^2)]^2}$$

$$\xi_2 = \frac{6\mathrm{tr}(\mathbf{\Sigma}^4)+2[\mathrm{tr}(\mathbf{\Sigma}^2)]^2}{[\mathrm{tr}(\mathbf{\Sigma}^2)]^2} + 6\gamma\frac{\mathrm{tr}(\mathbf{A}^2\odot\mathbf{A}^2)}{[\mathrm{tr}(\mathbf{\Sigma}^2)]^2} + \gamma^2\frac{\sum_{i=1}^p\sum_{j=1}^p A_{ij}^4}{[\mathrm{tr}(\mathbf{\Sigma}^2)]^2},$$

where the ratios in ξ_1 and ξ_2 involving γ vanish under Assumption 2.2. For the rump parts of the two variances, $0<\xi_c<\infty$, $c=1,2$, for any p, satisfying the condition of asymptotic normality of U_n (see (13) and the discussion around it), where

$$\mathrm{Var}(U_n) = \frac{4}{n(n-1)[\mathrm{tr}(\mathbf{\Sigma}^2)]^2}\big((2n-1)\mathrm{tr}(\mathbf{\Sigma}^4)+[\mathrm{tr}(\mathbf{\Sigma}^2)]^2\big).$$

Then, with $n, p \to \infty$, this leads to the proof of Eqs. (28) and (29). Finally, under H_{0i}, $\mathrm{Var}(U_n)$ clearly reduces to $\frac{4}{n(n-1)}$, as $p \to \infty$, so that, with both $n, p \to \infty$, $T_i \xrightarrow{\mathcal{D}} N(0,4)$, $i=1,2$, as required to complete the proof. □

3. Applications

The test statistics, T_1 and T_2, can be applied in a variety of research questions where the dimension of the multivariate vector exceeds the number of such vectors. Typical, but by no means exhaustive, of these fields are Microarray, Chemometrics, Genetics, even Agriculture and Engineering. Some specific examples from clinical experiments with moderate dimension exceeding the sample size are discussed in Ahmad (2008); see also Ahmad et al (2008). For similar applications with very large dimensions, see, for example, Izenman (2008).

For illustrative purposes, we here show the application of the test statistics for two real data sets. The first example in the following shows an application in Microarray analysis, and the other in Chemometrics. The data sets used for these examples are discussed in Izenman (2008).

Example 1: Cancer data The data used for this example consist of 62 colon tissue samples, 40 tumor and 22 normal, wherein 2000 gene expressions are recorded based on Affymetrix oligonucleotide array. This is a subset of the original data of 6500 expressions, as originally reported in Alon et al (1999), and further discussed in Izenman (2008, p. 19). With $n = 62$ and $p = 2000$, we compute the statistic, $T_1 = 30.72$ with p-value 0, which clues to a serious rejection of the data. Note that this data are also analyzed, for the same hypothesis, in Fisher et al. (2010) with the same p-value.

Example 2: PET data These data consist of $p = 286$ structural characteristics measured on $n = 21$ yarns, using spectroscopy. The original data (Swieringa et al., 1999) is for regression set up with $p = 286$ regressors used to predict the overall density of yarn which is the dependent variable. For further discussion of the data set and its use, see Izenman (2008, p. 129). For these data, we tested if the regressors are uncorrelated with each other, i.e., $H_0 : \mathbf{\Sigma} = \mathbf{I}$, where $\mathbf{\Sigma}$ refers to the 286×286 covariance matrix of regressors. We compute $T_2 = 10.95$ with p-value = 0, indicating that the regressors seem to be highly correlated.

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Appendix

A. Some basic results

Let $\mathbf{X}_k$ and $\mathbf{Y}_k$ be as defined in Sec. 2. Then, $A_k = \mathbf{X}_k'\mathbf{X}_k = \mathbf{Y}_k'\mathbf{A}\mathbf{Y}_k$, and $A_{kl} = \mathbf{X}_k'\mathbf{X}_l = \mathbf{Y}_k'\mathbf{A}\mathbf{Y}_l$, $k \neq l$. The following results on the moments of quadratic and bilinear forms will be needed to prove the main results.

Theorem A.1. *For A_k and A_{kl}, as defined above, we have*

$$E\big(A_k^2\big) = 2tr(\mathbf{\Sigma}^2) + [tr(\mathbf{\Sigma})]^2 + \gamma tr(\mathbf{A} \odot \mathbf{A}) \tag{30}$$

$$E\big(A_{kl}^4\big) = 6tr(\mathbf{\Sigma}^4) + 3[tr(\mathbf{\Sigma}^2)]^2 + 6\gamma tr(\mathbf{A}^2 \odot \mathbf{A}^2) + \gamma^2 \sum_{s=1}^{p}\sum_{t=1}^{p} A_{st}^4 \tag{31}$$

$$\begin{aligned} E\big(A_k A_l A_{kl}^2\big) &= 4tr(\mathbf{\Sigma}^4) + 4tr(\mathbf{\Sigma}^3)tr(\mathbf{\Sigma}) + [tr(\mathbf{\Sigma})]^2 tr(\mathbf{\Sigma}^2) + 2\gamma tr(\mathbf{\Sigma})tr(\mathbf{A}^2 \odot \mathbf{A}) \\ &\quad + 4\gamma tr(\mathbf{A}^3 \odot \mathbf{A}) + \gamma tr(\mathbf{A} \odot \mathbf{A}\mathbf{D}\mathbf{A}), \end{aligned} \tag{32}$$

where $\mathbf{D} = diag(\mathbf{A})$

Proof. The first two parts are trivial and their proofs will only be briefly sketched. Let $\mathbf{A} = (a_{st})$ with a_{ss} on its diagonal, $s, t = 1, \ldots p$. Then, for $A_k = \mathbf{X}_k'\mathbf{X}_k = \mathbf{Y}_k'\mathbf{A}\mathbf{Y}_k$, we have, under Assumption 2.1,

$$\begin{aligned}
\mathrm{E}(A_k^2) &= \mathrm{E}\left(\sum_{s=1}^{p}\sum_{t=1}^{p} a_{st}Y_{ks}Y_{kt}\right)^2 = \mathrm{E}\left(\sum_{s=1}^{p} a_{ss}^2 Y_{ks}^2 + \sum_{\substack{s=1 \\ s\neq t}}^{p}\sum_{t=1}^{p} a_{st}Y_{ks}Y_{kt}\right)^2 \\
&= (3+\gamma)\sum_{s=1}^{p} a_{ss}^2 + \sum_{\substack{s=1 \\ s\neq t}}^{p}\sum_{t=1}^{p} a_{ss}a_{tt} + 2\sum_{\substack{s=1 \\ s\neq t}}^{p}\sum_{t=1}^{p} a_{st}^2 \\
&= \gamma\sum_{s=1}^{p} a_{ss}^2 + \left(\sum_{s=1}^{p} a_{ss}\right)^2 + 2\sum_{s=1}^{p}\sum_{t=1}^{p} a_{st}^2 \\
&= \gamma\,\mathrm{tr}(\mathbf{A}\odot\mathbf{A}) + [\mathrm{tr}(\mathbf{A})]^2 + 2\mathrm{tr}(\mathbf{A}^2),
\end{aligned}$$

which immediately yields (30). We also note that, if we define another quadratic form $B_k = \mathbf{Y}_k'\mathbf{B}\mathbf{Y}_k$, then the following result can be similarly proved.

$$\mathrm{E}(A_kB_k) = \gamma\,\mathrm{tr}(\mathbf{A}\odot\mathbf{B}) + \mathrm{tr}(\mathbf{A})\mathrm{tr}(\mathbf{B}) + 2\mathrm{tr}(\mathbf{AB}). \tag{33}$$

For (31), write $A_{kl}^2 = \mathbf{X}_k'\mathbf{X}_l\mathbf{X}_k'\mathbf{X}_l = \mathbf{X}_k'\mathbf{X}_l\mathbf{X}_l'\mathbf{X}_k$, by symmetry. Now, let $\mathbf{X}_l\mathbf{X}_l' = \mathbf{M}$ so that $\mathbf{X}_l'\mathbf{X}_l = \mathrm{tr}(\mathbf{M})$. Then, the result follows by using (30) repeatedly on the conditional expectation

$$\mathrm{E}(A_{kl}^4) = \mathrm{E}(\mathbf{X}'\mathbf{M}\mathbf{X})^2 = \mathrm{E}[\mathrm{E}(\mathbf{X}_k'\mathbf{M}\mathbf{X}_k)^2|\mathbf{M}].$$

To prove (32), we follow the same notation and write

$$\begin{aligned}
\mathrm{E}(A_kA_lA_{kl}^2) &= \mathrm{E}(\mathbf{X}_k'\mathbf{X}_k\mathbf{X}_l'\mathbf{X}_l\mathbf{X}_k'\mathbf{X}_l\mathbf{X}_k'\mathbf{X}_l) = \mathrm{E}(\mathbf{X}_k'\mathbf{X}_k\mathbf{X}_l'\mathbf{X}_l\mathbf{X}_k'\mathbf{X}_l\mathbf{X}_l'\mathbf{X}_k) \\
&= \mathrm{E}\left[\mathrm{E}(\mathbf{X}_k'\mathbf{X}_k\mathrm{tr}(\mathbf{M})\mathbf{X}_k'\mathbf{M}\mathbf{X}_k|\mathbf{M})\right] \\
&= \mathrm{E}\left[\mathrm{tr}(\mathbf{M})\mathrm{E}(\mathbf{Y}_k'\mathbf{A}\mathbf{Y}_k\cdot\mathbf{Y}_k'\mathbf{B}'\mathbf{M}\mathbf{B}\mathbf{Y}_k|\mathbf{M})\right] \\
&= \mathrm{E}[\mathrm{tr}(\mathbf{M})\{2\mathrm{tr}(\mathbf{AB}'\mathbf{MB}) + \mathrm{tr}(\mathbf{A})\mathrm{tr}(\mathbf{B}'\mathbf{MB}) + \gamma\,\mathrm{tr}(\mathbf{A}\odot\mathbf{B}'\mathbf{MB})\}].
\end{aligned}$$

We deal with the terms inside brackets one by one. For the first term, we have

$$\begin{aligned}
\mathrm{E}[\mathrm{tr}(\mathbf{M})\mathrm{tr}(\mathbf{AB}'\mathbf{MB})] &= \mathrm{E}(\mathbf{X}_l'\mathbf{X}_l\cdot\mathbf{X}_l'\mathbf{BAB}'\mathbf{X}_l) = \mathrm{E}(\mathbf{Y}_l'\mathbf{A}\mathbf{Y}_l\cdot\mathbf{Y}_l'\mathbf{A}^3\mathbf{Y}_l) \\
&= 2\mathrm{tr}(\boldsymbol{\Sigma}^4) + \mathrm{tr}(\boldsymbol{\Sigma})\mathrm{tr}(\boldsymbol{\Sigma}^3) + \gamma\,\mathrm{tr}(\mathbf{A}\odot\mathbf{A}^3).
\end{aligned}$$

The second term expands to

$$\begin{aligned}
\mathrm{tr}(\mathbf{A})\mathrm{E}[\mathrm{tr}(\mathbf{M})\mathrm{tr}(\mathbf{B}'\mathbf{MB})] &= \mathrm{tr}(\mathbf{A})\mathrm{E}(\mathbf{X}_l'\mathbf{X}_l\cdot\mathbf{X}_l'\mathbf{BB}'\mathbf{X}_l) \\
&= \mathrm{tr}(\mathbf{A})\mathrm{E}(\mathbf{Y}_l'\mathbf{A}\mathbf{Y}_l\cdot\mathbf{Y}_l'\mathbf{A}^2\mathbf{Y}_l) \\
&= 2\mathrm{tr}(\boldsymbol{\Sigma})\mathrm{tr}(\boldsymbol{\Sigma}^3) + [\mathrm{tr}(\boldsymbol{\Sigma})]^2\mathrm{tr}(\boldsymbol{\Sigma}^2) + \gamma\,\mathrm{tr}(\boldsymbol{\Sigma})\mathrm{tr}(\mathbf{A}\odot\mathbf{A}^2).
\end{aligned}$$

For the last part, we use properties of Hadamard product and get

$$\begin{aligned}
\mathrm{E}[\mathrm{tr}(\mathbf{M})\mathrm{tr}(\mathbf{A}\odot\mathbf{B}'\mathbf{MB})] &= \mathrm{E}[A_l\mathrm{tr}(\mathbf{B}'\mathbf{MB}\odot\mathbf{A})] = \mathrm{E}[A_l\mathrm{tr}\{(\mathbf{B}'\mathbf{MB}\odot\mathbf{A})\cdot\mathbf{I}\}] \\
&= \mathrm{E}[A_l\mathrm{tr}\{\mathbf{B}'\mathbf{MB}(\mathbf{A}\odot\mathbf{I})\}] = \mathrm{E}[A_l\mathrm{tr}(\mathbf{B}'\mathbf{MBD})]
\end{aligned}$$

$$= \mathrm{E}[\mathbf{X}_l'\mathbf{X}_l \cdot \mathbf{X}_l'\mathbf{BDB}'\mathbf{X}_l)] = \mathrm{E}[\mathbf{Y}_l'\mathbf{AY}_l \cdot \mathbf{Y}_l'\mathbf{ADAY}_l)]$$
$$= 2\mathrm{tr}(\mathbf{DA}^3) + \mathrm{tr}(\boldsymbol{\Sigma})\mathrm{tr}(\mathbf{DA}^2) + \gamma\mathrm{tr}(\mathbf{A} \odot \mathbf{ADA})$$
$$= 2\mathrm{tr}(\mathbf{A} \odot \mathbf{A}^3) + \mathrm{tr}(\boldsymbol{\Sigma})\mathrm{tr}(\mathbf{A} \odot \mathbf{A}^2) + \gamma\mathrm{tr}(\mathbf{A} \odot \mathbf{ADA}).$$

Combining the results, we get the required identity. □

B. Proof of Lemma 2.1

The unbiasedness of the three estimators follows immediately due to independence of the vectors $\mathbf{X}_k$. For variances, we use Eq. (12). We first note that, for $A_k = \mathbf{X}_k'\mathbf{X}_k = \mathbf{Y}_k'\mathbf{AY}_k$, Eq. (30) gives, under Assumption 2.2,

$$\mathrm{Var}\left(\frac{A_k}{\mathrm{tr}(\boldsymbol{\Sigma})}\right) = \frac{2\mathrm{tr}(\boldsymbol{\Sigma}^2) + \gamma\mathrm{tr}(\mathbf{A} \odot \mathbf{A})}{[\mathrm{tr}(\boldsymbol{\Sigma})]^2} \leq 2 + \gamma\frac{\mathrm{tr}(\mathbf{A} \odot \mathbf{A})}{[\mathrm{tr}(\boldsymbol{\Sigma})]^2} \approx 2, \tag{34}$$

Now, for E_1, we have $h_1(\mathbf{X}_k) = \frac{1}{\mathrm{tr}(\boldsymbol{\Sigma})}A_k$, for $c = 1$, so that $\xi_1 \leq 2$. This proves the result for E_1, and also for E_2, again using independence and since $k \neq l$. For E_3, we have, for Eq. (12), $h_1(\mathbf{X_k}) = \frac{1}{\mathrm{tr}(\boldsymbol{\Sigma}^2)}\mathbf{X}_k'\mathbf{AX}_k = \frac{1}{\mathrm{tr}(\boldsymbol{\Sigma}^2)}\mathbf{Y}_k'\mathbf{A}^2\mathbf{Y}_k$ with $c = 1$. Then, replacing $\mathbf{A}$ with $\mathbf{A}^2$, Eq. (34) gives $\xi_1 \leq 2$, under Assumption 2.2. Further, with $c = 2$, $h_2(\mathbf{X_k}, \mathbf{X}_l) = \frac{1}{\mathrm{tr}(\boldsymbol{\Sigma}^2)}(\mathbf{X}_k'\mathbf{X}_k)^2 = \frac{1}{\mathrm{tr}(\boldsymbol{\Sigma}^2)}(\mathbf{Y}_k'\mathbf{AY}_k)^2$, and Eq. (31) gives $\xi_2 \leq 8$, again under Assumption 2.2, which proves the result for E_3.

For covariances, we use Eq. (14). Then, $\mathrm{Cov}(E_1, E_2) = \frac{1}{n}2\xi_1$, with

$$\xi_1 = \mathrm{Cov}(A_k, A_k\mathrm{tr}(\boldsymbol{\Sigma})) = \mathrm{tr}(\boldsymbol{\Sigma})\mathrm{Var}(A_k) = \mathrm{tr}(\boldsymbol{\Sigma})[2\mathrm{tr}(\boldsymbol{\Sigma}^2) + \gamma\mathrm{tr}(\mathbf{A} \odot \mathbf{A})].$$

Similarly, $\mathrm{Cov}(E_1, E_3) = \frac{1}{n}2\xi_1$, where

$$\xi_1 = \mathrm{Cov}(A_k, \mathbf{X}_k'\boldsymbol{\Sigma}\mathbf{X}_k) = 2\mathrm{tr}(\boldsymbol{\Sigma}^3) + \gamma\mathrm{tr}(\mathbf{A}^2 \odot \mathbf{A}).$$

Finally,

$$\mathrm{Cov}(E_2, E_3) = \frac{2}{n(n-1)}[2(n-2)\xi_1 + \xi_2],$$

where

$$\xi_1 = \mathrm{Cov}(A_k\mathrm{tr}(\boldsymbol{\Sigma}), \mathbf{X}_k'\boldsymbol{\Sigma}\mathbf{X}_k) = \mathrm{tr}(\boldsymbol{\Sigma})[2\mathrm{tr}(\boldsymbol{\Sigma}^3) + \gamma\mathrm{tr}(\mathbf{A}^2 \odot \mathbf{A})].$$

Further, $\xi_2 = \mathrm{Cov}(A_k, A_l, A_{kl}^2) = \mathrm{E}(A_kA_lA_{kl}^2) - [\mathrm{tr}(\boldsymbol{\Sigma})]^2\mathrm{tr}(\boldsymbol{\Sigma}^2)$, so that, from Eq. (32), we have

$$\xi_2 = 4\mathrm{tr}(\boldsymbol{\Sigma}^4) + 4\mathrm{tr}(\boldsymbol{\Sigma})\mathrm{tr}(\boldsymbol{\Sigma}^3) + 4\gamma\mathrm{tr}(\mathbf{A} \odot \mathbf{A}^3)$$
$$+ 2\gamma\mathrm{tr}(\boldsymbol{\Sigma})\mathrm{tr}(\mathbf{A} \odot \mathbf{A}^2) + \gamma\mathrm{tr}(\mathbf{A} \odot \mathbf{ADA}).$$

Substituting ξ_1 and ξ_2, we get the required covariance.

Note that, under normality, $\gamma = 0$ and the results of Theorem 2.1 and Lemma 2.1 immediately reduce to those when $\mathbf{X}_k \sim \mathcal{N}(\mathbf{0}, \boldsymbol{\Sigma})$.